Machine Learning

23-03-26

Step 5: ML Framework → Model training

1) Identify type of learning for given task
   A) Supervised B) Unsupervised

2) Identify type of model / algorithm

A) Supervised

i) regression
- KNN regr
- Linear regr
- Decision Tree regr
- Random Forest
- neural network
- SVM

ii) classification
- KNN
- Logistic regr
- Naive bayes
- Decision Tree
- Random Forest
- Neural network

B) Unsupervised

i) clustering
- Kmeans
- DBSCAN
- Agglomorative

etc

ii) dimensionality reduction
- PCA

We try to find the <u>best possible</u> algorithm for given task from all available options.

best model → most accurate model

input → $\boxed{\text{model}}$ → target output
↓
prediction

- during model training we try to teach model the task that we want to automate using ML

- Once model training is successful, we use the trained model to make predictions on new examples.

- There are multiple models/ algos in ML
- before using any model, we need to understand its below properties
  - how training and testing happens
  - mathematical details and computations
  - how data is prepared for training / testing
  - is this model suitable for given task?
  - we must know reasons for good/ bad model performance
  - how given model converts input to output

Important aspects of any ML model

- Hypothesis function
  - tells how model converts input to output

$$\text{target} = f(\text{input})$$

$$y = f(x)$$

↓

hypothesis function

- Loss / cost function
  - this tells how model corrects its mistakes

- Performance metrics
  - evaluates model performance
  - tells if model training was successful or not

KNN Algorithm

- K Nearest Neighbour algorithm
- its a distance based algorithm
- supervised ML model
- used for regression and classification
- non-parametric model

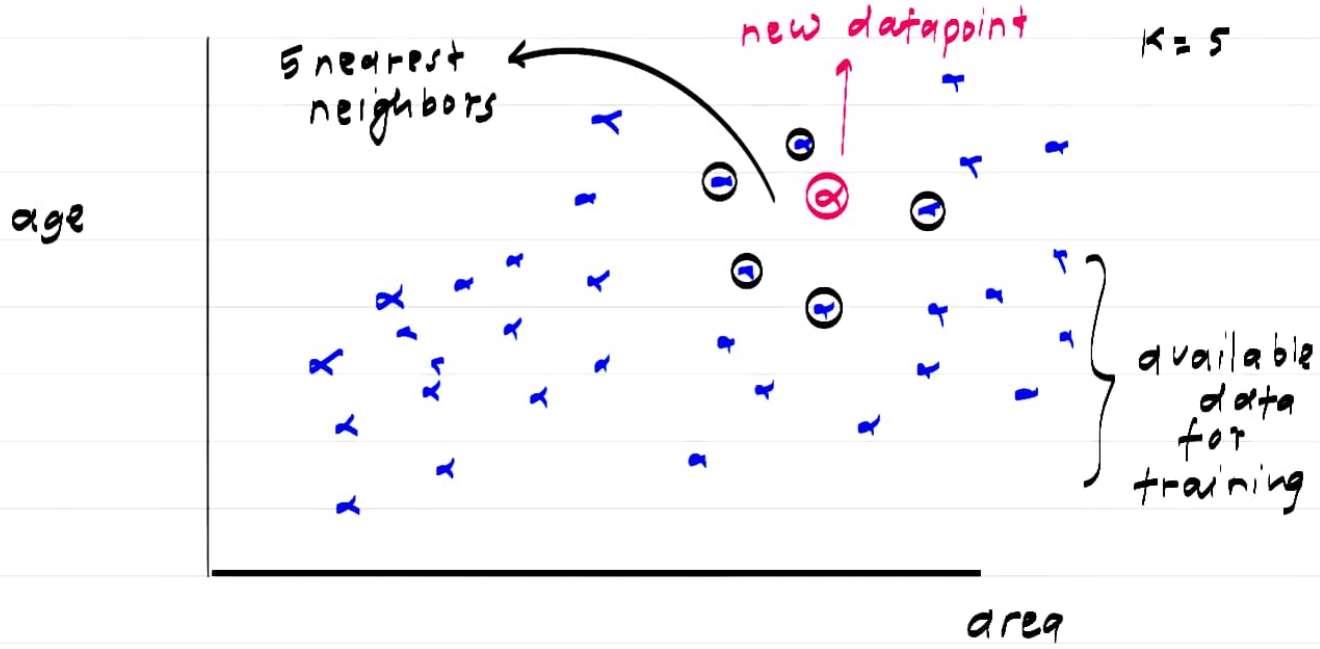
KNN Regression

eg. house price prediction

area age price → continuous

area
age → [KNN Regr] → price

scatterplot of area vs age



1) Divide available data into train and test subsets
   train data - training
   test data - testing / evaluation

2) Store all training datapoints in the model

3) set the value of K as integer
   K is no. of nearest neighbors to use

4) Evaluate model performance on train data

5) Inference - test model on testing data

Train test split of data

Available data → 1000 rows

train	test
800 rows	200 rows
↓	↓
training model	test / evaluation

First we train the model on train data

- we compute model's training accuracy

if model's train accuracy is good, we can say that model has understood train data

- high train accuracy doesn't mean successful model training

After good training accuracy, we evaluate model on test data

- we'll compute test accuracy

if model's test accuracy is also good, it means that model training is successful.

train acc	test acc	model status
good	good	good model
good	bad	not a good model

generally used train : test ratios

8 : 2 → 80% train & 20% test

85 : 15 → 85% train & 15% test

When we do train-test split, we need to make sure that the data distributions remain same in both train and test set.

eg. house price prediction

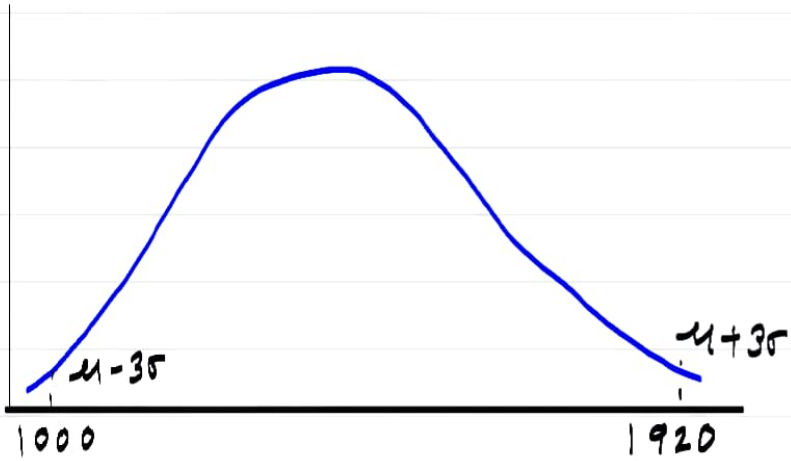
area	age	rooms	city	price

available data | 1000 rows < 800 train, 200 test

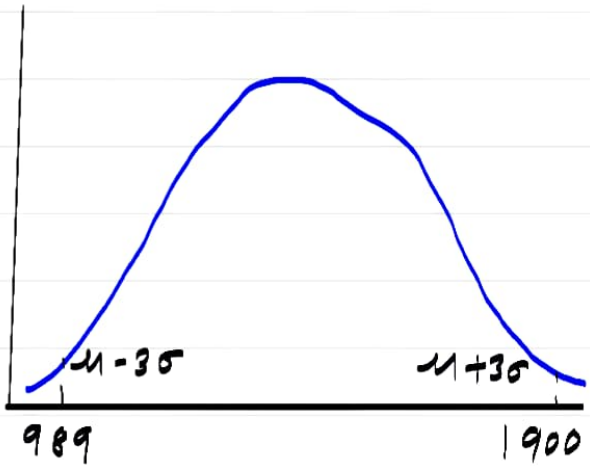
1000 rows → distribution of area

* dist of area is nearly same for available data, train, test data
- this is a good data split

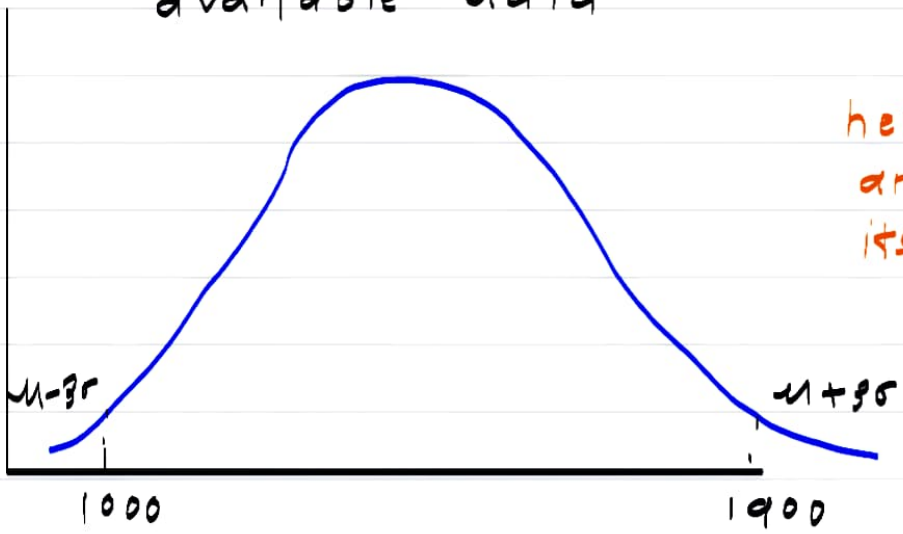
800 rows - train



200 rows - test

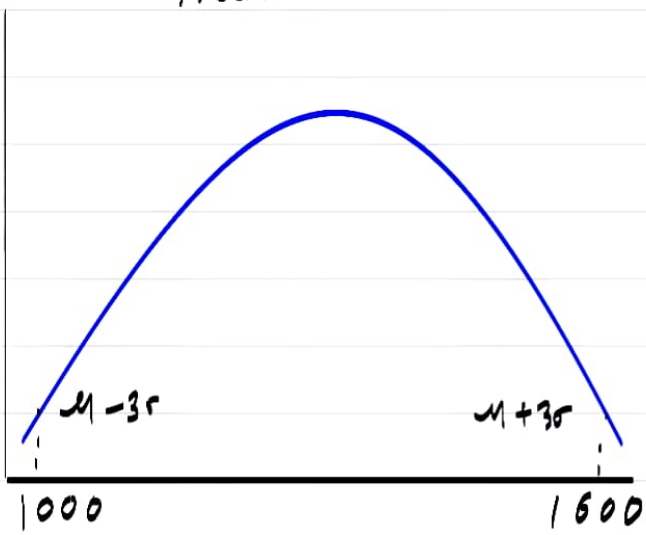
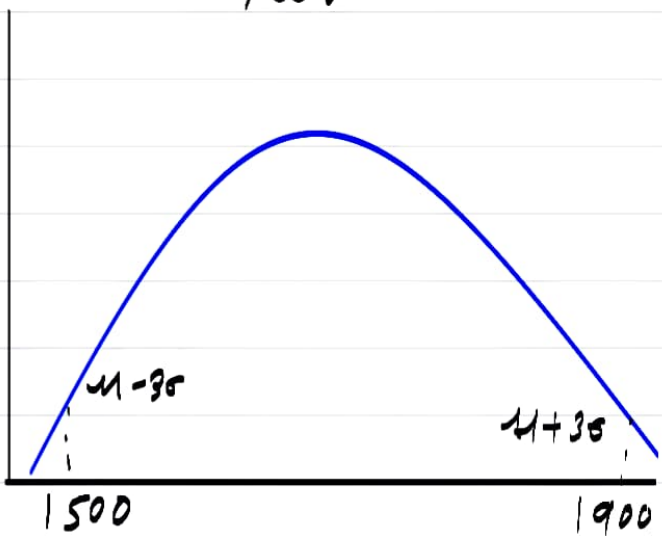


available data



here distributions are not similar hence its a bad train-test split

train



test